

João Victor Mendes

Production Support Engineer (PSE) / SRE | Cloud · Observability · Database Reliability

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PROFESSIONAL SUMMARY

Production Support Engineer (PSE) and SRE supporting critical distributed platforms (FinTech, AI, and SaaS). Hands-on expertise in relational database internals and saturation diagnostics (AWS Aurora MySQL, PostgreSQL), production P1/P2 incident response, detection engineering focused on MTTD reduction, and root-cause post-mortems.

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (Aurora MySQL/RDS, EKS, CloudWatch, SSM), Kubernetes (Karpenter, ArgoCD, Topology Spread Constraints), Docker, Linux (systemd, boot/network troubleshooting), Nginx.
- **Observability & SRE:** New Relic (NRQL, NerdGraph API, baseline/alerting), OpenTelemetry, Incident Response (P1/P2), Post-Mortem / RCA (ITIL), Detection Engineering & Runbooks.
- **Databases:** AWS Aurora MySQL (Performance Insights, connection pooling, parameter groups), PostgreSQL (WAL, replication slots, bloat, unlogged tables, `pg_class`, `pg_attribute`, `pg_toast` catalogs, EXPLAIN ANALYZE), Firebird.
- **Security & Access:** `pg_hba` access hardening (SCRAM-SHA-256 vs. MD5, scoped IP ranges), audit log analysis, secret management (SSM Parameter Store SecureString).
- **IaC, Automation & Testing:** Terraform, Terragrunt, Python, Bash, PowerShell, SQL, PL/pgSQL, Playwright, Git.
- **Languages:** English (B2 Certified, BRASAS — daily technical documentation and incident response), Portuguese (Native).

PROFESSIONAL EXPERIENCE

Monest | Credit Recovery FinTech & AI Platform Aug 2026 – Present
Production Support Engineer (PSE) / SRE Remote, Brazil

Engineering reliability and observability for a distributed cloud platform powering AI agents, VoIP dialers, and microservices.

- **Database Saturation Diagnostics & Capacity (Aurora MySQL):** Diagnosed the root cause of CPU and connection saturation on a production Aurora cluster, separating query-concentrated load in Performance Insights (>90% of writer load) from connection-pool amplification (pools totaled ~15% above cluster ceiling); supported prioritizing RDS Proxy.
- **Detection Engineering & Alerting Governance (New Relic / Terraform):** Audited 400+ alert conditions via GraphQL, isolating an unsegmented policy driving ~40% of incident volume (~4,900/month) and a baseline pattern that could suppress notifications on sparse workloads; authored Terraform policies calibrated on 7-day baselines, recalibrating load alarms breaching over a third of the time that suppressed incident pages.
- **External Integration Root Cause (API / Observability):** Diagnosed intermittent failures in an external-service integration: schema validation inside the `try` block masked HTTP 200 responses as 500 errors; comparing logged errors against genuinely failed HTTP calls separated hypotheses, dropping errors from 9,428 to 3 after the owning team's schema fix.
- **Data Pipeline Incident Response (Airflow / Docker / SSM):** Led post-mortem for a pipeline outage from disk exhaustion; bypassed SSH blockers via SSM without admin, validated with historical data that an NRQL deadman alert would have cut MTTD from ~2h to ~10 min (surfacing failure 1h40 before human triage), and fixed an IaC disk rule where mount averaging masked ~99.9% root saturation.
- **Scheduler Outage & Host Hardening (Systemd / SSM / Linux):** Traced root cause of a prolonged job-scheduler outage: monitoring checked proxy availability, not execution; during a controlled reboot, diagnosed boot races and missing restart policies, extended resilient systemd units and configured Slack failure alerts via `OnFailure`, with credentials in SSM Parameter Store.
- **K8s Resiliency (EKS / Karpenter / ArgoCD):** Strengthened cross-node replica distribution for the AI observability stack via GitOps/ArgoCD with strict `TopologySpreadConstraints (DoNotSchedule)`, preventing Karpenter single-node bin-packing after Spot node interruption.

Direct L2/L3 support for 500+ corporate environments (1,500+ end-users). Maintained full independence on solo rotations, resolving P1/P2 incidents without escalation through deep structural analysis.

- **WAL Retention Investigation & Guardrails:** Traced recurring fleet-wide disk exhaustion to orphaned slots in `pg_replication_slots` holding WAL segments for a decommissioned logical replication subscriber; built PowerShell automation detecting and dropping inactive slots without an active consumer (>2GB retained or stalling LSN progression), and enforced server-side `max_slot_wal_keep_size` on PG13+, eliminating recurring disk-full incidents.
- **Disaster Recovery & Access Hardening:** Parallelized cluster recovery across 30 databases (cutting downtime from hours to <5 min by automating `postmaster.pid` cleanup and startup WAL validation, removing human error under pressure); resolved client access lockouts in `pg_hba.conf` by limiting MD5 permissions to scoped IP ranges for legacy clients while enforcing SCRAM-SHA-256 across the rest of the fleet.
- **Bloat Diagnostics & Restore Optimization:** Diagnosed slow restore reports by auditing `pg_class`, `pg_attribute`, and TOAST tables; traced 1GB/month bloat to encrypted file paths serialized across thousands of characters per audit log entry, and accelerated restores by 60% (17 min down to 6:58) via parallelized `pg_restore -j 4`.
- **70M-Record ETL Pipeline Engineering:** Designed autonomous pipeline across 15GB+ government dataset (70M+ records) in PL/pgSQL and PowerShell, reducing processing time by 60% using unlogged tables to minimize write I/O overhead and dynamic SQL; implemented automated corporate tax classification and XML invoice auditing with staging in PostgreSQL.
- **Protected \$100k USD (R\$500k) in live inventory (zero data loss)** by reverse engineering an undocumented legacy database (Firebird) with no stored state; reprocessed 20 years of transaction logs to reconstruct inventory with full consistency, delivered the same day the schema was first accessed.
- **Reliability Culture, Mentorship & AI-Assisted Triage:** Authored 20+ video walkthroughs and ITIL runbooks adopted as team standards, converting daily junior escalations into independent case resolution without senior intervention; engineered LLM-assisted triage injecting database context and SEFAZ fiscal rules, reducing diagnosis time from 30+ to <5 min.

Operated Cloud infrastructure and web operations for 70+ corporate clients (AWS Route 53, Registro.br, Locaweb).

- **Cloud Operations & Incident Resolution:** Managed corporate DNS, WAF, and zones across AWS Route 53 and CloudFront for 70+ domains; served as international liaison in 15+ complex migrations with external registrars (GoDaddy, AWS, Namecheap) and isolated propagation/caching failures.
- **Security Incident Investigation:** Audited Git histories and access logs to identify security breaches; revoked compromised sessions and rotated credentials across 10+ corporate environments, preventing incident recurrence.

PRODUCTION ENGINEERING PROJECTS

- Automated multi-source collection (Copel, ANA, ONS) integrating basin telemetric stations with immutable raw payloads; Python API behind Nginx and DEM geoprocessing with caching and singleflight under concurrency.
- 120+ immutable releases with symlink rollback, systemd service hardening with memory limits (MemoryMax), health and readiness endpoints, backups, and Playwright test suite.
- Private experimental forecasting with immutable emission archives and prospective scorecard evaluation, isolated from public product.

EDUCATION & PEDIGREE